Initials:_____

ANSWER KEY

BME 105: Genetics in the Genomics Era
Spring 2019: Midterm 1

Name: _____ANSWER KEY_____

Student ID:_____

Before starting, write your initials on the top of each page.

You can use the back-side of the pages for scratch, but we will not grade answers written on the back-side of the page.

Section	Points
Part I: One, two, or three-word answers and T/F	/26
Part II: Multiple choice	/24
Part III: Short answer	/20
Part IV: Extended calculation	/30
Total Points	**/100**

Second Letter

1st letter	U		C		A		G		3rd letter
U	UUU UUC UUA UUG	Phe Leu	UCU UCC UCA UCG	Ser	UAU UAC UAA UAG	Tyr Stop Stop	UGU UGC UGA UGG	Cys Stop Trp	U C A G
C	CUU CUC CUA CUG	Leu	CCU CCC CCA CCG	Pro	CAU CAC CAA CAG	His Gln	CGU CGC CGA CGG	Arg	U C A G
A	AUU AUC AUA AUG	Ile Met	ACU ACC ACA ACG	Thr	AAU AAC AAA AAG	Asn Lys	AGU AGC AGA AGG	Ser Arg	U C A G
G	GUU GUC GUA GUG	Val	GCU GCC GCA GCG	Ala	GAU GAC GAA GAG	Asp Glu	GGU GGC GGA GGG	Gly	U C A G

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Part I: One, two, or three-word answers and True/False

In the space provided next to each definition, clearly write the appropriate term.

1. _____product rule_____ if two or more events are independent, you can use this rule to determine the probability of the events occurring together

2. _____F1_____ The first offspring from the parents

3. _____autosomal_____ a gene that lies on any chromosome except the sex chromosomes

4. _____testcross_____ this type of cross is performed when trying to determine linkage

5. single nucleotide polymorphism the abbreviation is SNP

6. _____hemizygous_____ males are considered this for every gene on the X chromosome, since they only have one copy.

7. _____exome_____ a type of sequencing where only protein-coding exons are sequenced

8. _____anaphase II_____ the phase of meiosis where sister chromatids are pulled apart

9. ____monohybrid cross_____ a cross between individuals, both heterozygous for one gene

10. _____haplotype_____a group of alleles that are inherited together on the same chromosome, due to linkage

11. Indicate if the following statements are True or False:
    a. T F Homologous chromosomes are not necessarily identical.

    T

    b. T F Independent assortment of two unlinked heterozygous loci can generate recombinant gametes.

    T

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12. News articles often describe genetic concepts inaccurately. In a recently published article from *People* magazine titled "Woman Gest Preventive Double Mastectomy and Then Learns the DNA Testing Was Wrong: 'Just Devastating'" (March 27, 2019), the first sentence says, "Nearly a decade after getting a preventative double mastectomy after testing showed she had the BRCA gene mutation, a Missouri woman learned that she did not actually have the gene." If I were to edit this sentence, I would change it to:

    "Nearly a decade after getting a preventative double mastectomy after testing showed she had a BRCA gene mutation, a Missouri woman learned that she did not actually have _____."

    There are many different mutations in the BRCA gene that can increase risk of breast cancer. Since there are many different mutations, instead of referring to "the" BRCA gene mutation, I would have changed it to "a" BRCA gene mutation.

    What would you replace at the end of the sentence to make it more accurate?

    "the allele" or "the mutation". There may be other possible correct answers

Part II: Multiple choice (circle your answer. If there is more than one correct answer, circle all correct answers)

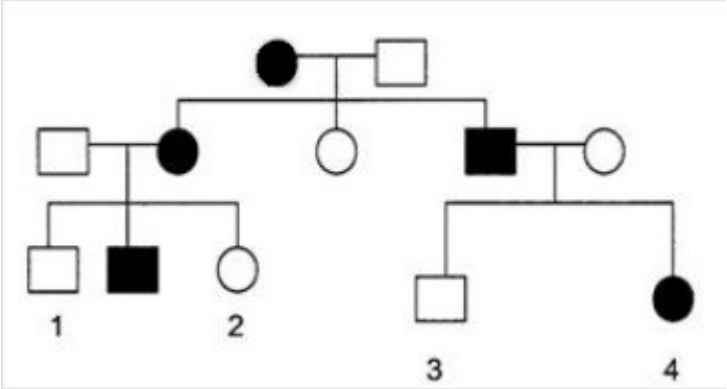
1. In *Drosophila virilis*, somatic cell nuclei contain 12 chromosomes while sperm nuclei contain only 6 chromosomes. What does *N* equal for this species?
    a. 3
    b. 6
    c. 12
    d. 24
    e. 6 or 12, depending on cell type

    b

2. During which of the following stages of the cell cycle would a chromosome consist of only a single chromatid?
    a. G_1
    b. G_2
    c. mitotic prophase
    d. mitotic metaphase

    a

3. Below is a pedigree of a human genetic disease in which solid color indicates affected individuals.



What is the likely mode of inheritance?

a. Dominant
b. Recessive
c. Cannot be determined

a

4. If we cross 2 plants that are both heterozygous at three unlinked loci, what fraction of the progeny will be homozygous for all three recessive alleles?

a. 1/16

b. 1/64

c. 3/64

d. 1/256

e. 12/256

b

5. In corn, having ligules (*L)* is dominant to liguleless (*l*), and green leaves (*G*) is dominant to white leaves (*g*). If a cross was performed with a dihybrid plan and a ligueless and white plant, what proportion of the progeny would be green and liguleless?
    a. 1/16
    b. 1/8
    c. 1/4
    d. 1/2
    e. Cannot be determined

    c

6. If a pure-breeding liguleless plant with green leaves is crossed to pure-breeding plant with ligules and white leaves, predict the proportion of F2 progeny with the genotype *Ll gg*.
    a. 0
    b. 1/16
    c. 1/8
    d. 1/4
    e. 1/2
    f. Cannot be determined

    c

7. Among the dihybrid crosses below, which will produce offspring with a 1:1:1:1 genotypic ratio?
    a. *AA BB* × *aa bb*
    b. *Aa Bb* × *Aa Bb*
    c. *Aa Bb* × *aa bb*
    d. *Aa BB* × *aa BB*
    e. *AA bb* × *aa BB*

    c

8. Nondisjunction can occur at either the first or second division of meiosis. XYY individuals would most likely arise from nondisjunction at the _____ meiotic division in the _____. Note that in meiosis, the X and Y chromosomes can pair due to a pseudoautosomal region.

   a. first; mother
   b. second; mother
   c. first; father
   d. second; father

   d

9. Hemophilia is caused by an X-linked recessive mutation in humans. If a woman whose paternal uncle was a hemophiliac marries a man whose brother is also a hemophiliac, what is the probability that their first child will have hemophilia? (Assume that no other cases of hemophilia and no hidden carriers exist in the pedigree.)
   a. 0
   b. 1/4
   c. 1/2
   d. 1/8
   e. 1

   a

10. The *R* and *S* genes are linked and 10 map units apart. In the cross *R s / r S* × *r s / r s* what fraction of the progeny will be *R S / r s*?

   a. 5%
   b. 10%
   c. 25%
   d. 40%
   e. 45%

   a

11. Individuals heterozygous for the *RB*+ and *RB*− alleles can develop tumors as a result of
    a. a mitotic crossover that leads to homozygosity for *RB*+ in some cells and *RB*− in other cells.
    b. a somatic mutation in the *RB*+ allele that leads to homozygosity for *RB*−.
    c. a somatic mutation in the *RB*− allele that leads to homozygosity for *RB*+.
    d. the fact that *RB*− is dominant to *RB*+.

    a, b

12. A female mouse from a true-breeding wild-type strain was crossed to a male mouse with apricot eyes (*ap*) and grey body (*gy*). The F_1 mice were wild-type for both traits. When the F_1 were interbred, the F_2 were distributed as follows:

Females

all wild type	200

Males

wild type	91
apricot	11
grey	9
apricot, grey	89

Which of the following statements is correct?
a. *ap* and *gy* are unlinked
b. *ap* and *gy* are linked on an autosome and 10 map units apart
c. *ap* and *gy* are linked on an autosome and 20 map units apart
d. *ap* and *gy* are X-linked and 10 map units apart
e. *ap* and *gy* are X-linked and 20 map units apart

d

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Part III: Short answer

1. In frogs, green skin is determined by gene *B*. The wild-type sequence of the 5′ end of the RNA-like strand of the entire first exon is shown below. The gene encodes an enzyme that functions to convert a brown compound into a green pigment in the skin.

   5′ ACTCAAGCACAGGTCGCATAAATGTTCCTGTTATTTGG… 3′

   What are the first three amino acids encoded by the wild-type sequence of gene *B*?

   Met-Phe-Leu. The first three codons are highlighted above. Protein sequence always starts with Met.

2. In humans, the genes for red-green color blindness (*R* = normal, *r* = color blind) and hemophilia A (*H* = normal, *h* = hemophilia) are both X-linked and only 3 map units apart.
   a. Suppose a woman has four sons, and two are color blind but have normal blood clotting and two have hemophilia but normal color vision. What is the probable genotype of the woman? Make sure the genotype is written correctly for linked genes.

      Hr/hR

   b. A woman whose mother is color blind and whose father has hemophilia A is pregnant with a boy. If the alleles for color blindness and hemophilia A are rare in the population, what is the probability that the baby will have normal vision and normal blood clotting?

      0.015

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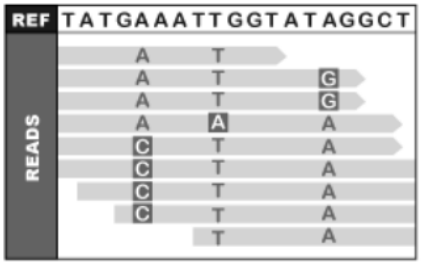
3. A child has a rare phenotype that no one has ever seen before. As the child's physician, you think the child has a very very rare disease. To help figure out a genetic cause of the disease you are considering two possible options, whole-genome sequencing or SNP microarray. For each option, explain why the method **would** or **would not** be the best way to figure out the genetic cause of the disease.
    a. Whole-genome sequencing
        i. Circle one: **would** or **would not**
        ii. Why?

The disease is thought to be caused by a rare disease, so it is unlikely to know what gene or mutation is causing the disease. Whole-genome sequencing would allow you to sequence the entire genome and find all genes that are likely mutated in the child.

    b. SNP microarray
        i. Circle one: **would** or **would not**
        ii. Why?

Since this is a rare disease, it is unlikely that you will know what the genetic cause of the disease is. For SNP arrays, you have to know what SNPs in the human genome you are genotyping for. These are typically common SNPs in a population or SNPs that are already known to be associated with disease.

4. Through high-throughput sequencing, you would like to determine the haplotype of an individual. What is shown below are the read alignments against the reference genome. Positions that differ from the reference in 1 or more reads are highlighted.



a. One of the reads is most likely caused by an error, what is the most likely cause of this error?

Sequencing error or **PCR artifact**. There is only one read that supports the "A" at the 2nd variant position. The error could have been introduced in PCR due to polymerase error or it could have occurred in the actual sequencer. Because of the local sequence context shown, it is unlikely due to mapping errors of the reads to the reference sequence.

b. What are the two most likely haplotypes for this individual?

The answer can be given as the full sequence, including invariant positions, or just the variants.

Full sequence:
Chromosome 1: TATG<u>A</u>AATTGGTAT<u>G</u>GGCT
Chromosome 2: TATG<u>C</u>AATTGGTAT<u>A</u>GGCT

Just variants:
Haplotype 1: AG
Haplotype 2: CA

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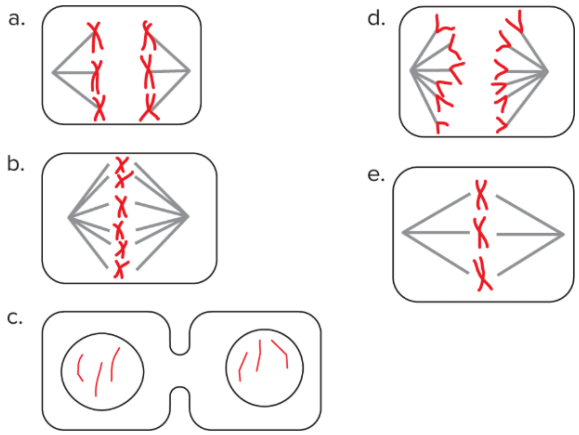
5. The five cells shown in figures a-e are all from the same individual.



Which stage of the mitotic or meiotic cycle is shown in **e**? Be as specific as possible.

Metaphase II of Meiosis II

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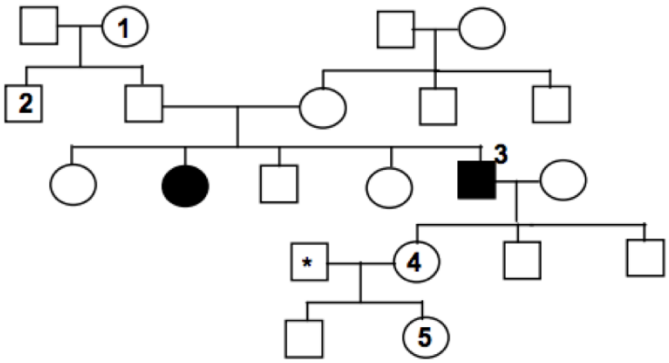
Part IV: Extended calculations (show your work for partial credit, show pedigrees and genotypes, punnett squares, etc.)

1. The following pedigtree shows a family affected by a specific disease. Assume that the individuals marked by an asterisk (*) do not carry any allele associated with the affected phenotype and that no other mutation spontaneously occurs.



a. Write all possible genotypes of the following individuals in the pedigree. Use the uppercase "A" for the allele associated with the dominant allele and a lower case "a" for the allele associated with the recessive allele.
   i. Individual #1

      AA or Aa

   ii. Individual #2

      AA or Aa

   iii. Individual #4

      Aa

b. What is the probability that Individual #5 is a carrier?

   Individual #4 is a carrier and the individual with asterisk does not carry the alleles, therefore the chance that individual is a carrier is 1/2 (the probability that individual #4 will pass on a recessive allele).

c. Through further investigation, Individual #2 discovers that his parents are both heterozygous. If Individual #2 has a child with an affected female, what is the probability that their child would be affected?

Probability that the child is affected is conditioned on the probability that individual #2 is a carrier. The prompt tells you that the parents are heterozygous.

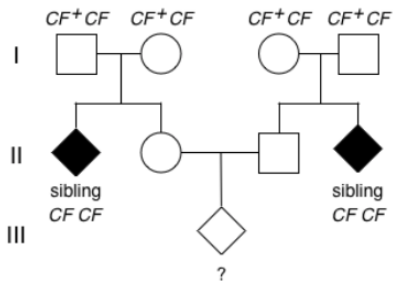
P (Child is aa and #2 is Aa) = P (Child is aa given that #2 is Aa) * P (#2 is Aa)

= 1/2 * P(#2 is Aa)

Since you know #2 is unaffected and the parents are both heterozygous, then the chance of #2 being a carrier is 2/3

= 1/2 * 2/3 = 2/6 = **1/3**

2. A young couple went to see a genetic counselor because each had a sibling with cystic fibrosis. Cystic fibrosis is a recessive disease, and neither member of the couple nor any of their four parents is affected. For the normal allele, use "CF+" and for the recessive allele use "CF".
   a. Draw the pedigree. Indicate genotypes, if known.



   b. What are the chances that their child will have cystic fibrosis?

Probability that III-1 will have cystic fibrosis is the probability that the child will be CF CF, but this is conditional on the parents being carriers (CF CF+)

P (III-1 is CF CF and <u>II-2 is CF CF+ and II-3 is CF CF+)</u>

What is underlined is the condition that III-1 is dependent on, so we use conditional probability to rewrite this as

= P (III-1 is CF CF **given** <u>II-2 is CF CF+ and II-3 is CF CF+)</u> * P (<u>II-2 is CF CF+ and II-3 is CF CF+)</u>

= 1/4 * P(II-2 is CF CF+ **and** II-3 is CF CF+)

The P(II-2 is CF CF+) is independent of P(II-3 is CF CF+), so we can use the product rule

= 1/4 * P(II-2 is CF CF+) * P(II-3 is CF CF+)

= 1/4 * 2/3 * 2/3 = **1/9**